WASTEWISE-ML · STEM CHECK

WasteWise-ml STEM check

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DOCUMENT CONTROL

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Author	Amish Chadha	License	MIT
Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-25	Amish Chadha	First STEM check (kit 1.8 and 1.9.1 upgrade): 31 claims across both concept options; four failed details fixed in the design under Amish's standing instruction of 2026-10-23 (WML-DEC-001 F1 to F3 and F5, WML-DDR-005); independent review recorded: S2, S7, M2, M5, B2, C3, E4 restated, lens height proposed to Amish (D2)

WasteWise-ml STEM check

Verdict: open (eight rows to confirm). WasteWise-ml's working principle, an image classifier on a phone that names the material class and grade of one item photographed on a mat, sits within known science: no row fails, and 23 of its 31 claims are within known science. Option A, the current design, has 26 rows; option B, the tray detector, has five. Eight rows stay **Plausible, to confirm**. Four rest on field numbers that only TRL 4 work can supply: the dataset size (M4), the share of items answered (M5, 66 % against 70 %), option B's hazard recall in a crowded tray (OB5) and the photo-to-result time if a full 12 MP frame is decoded (M2). Three need a measurement on bought or made parts: legibility of the flat screen for a short user (B2), the sealed mat staying flat (C3) and the shutter button's cell life (E4). One needs Amish: the picture fits the mat only for a lens of 25.76 mm equivalent or longer (S2), and lowering the lens to 420 mm would fix that but changes his 440 mm decision (N3), so it is proposed to him as D2 in WML-DEC-001. `project.yaml` records `stem_check: open`.

The first pass found four details that failed with a fix that keeps the working principle (the fourth, the phone's placement in the tray, came from the independent review), and one borderline reach. Under Amish's standing instruction of 2026-10-23 ("whenever a STEM check fails propose and implement the best fix to the design and constructability"), each fix is now in the model, BOM, build plan, precis and requirements and recorded in the design decisions register (WML-DEC-001 F1 to F3 and F5) and in WML-DDR-005:

- **F1, reach and reading (B1, B2).** The phone lies flat at 1,215 mm. For the shortest users (the 5th percentile Indian woman, 1,356 mm tall) the screen sits about 106 mm above the shoulder, so tapping it needs about 59 degrees of upper arm elevation at about one scan a minute, at the 60 degree screen of ISO 11226; and the eye is only about 54 mm above the screen, 83 degrees off its normal, where a 10 mm icon subtends about 11 arcmin against the 16 arcmin minimum of ISO 9241-303. Fix: a Bluetooth shutter button at the bench edge takes the photo, the result fills the screen in the class colour with the bin number at least 40 mm tall (about 43 arcmin for that user), and the phone says the bin number aloud by default.
- **F2, the mat (C3).** The build plan painted and varnished the mat's top face only; hardboard swells up to 25 % in thickness through unsealed faces and edges. Fix: prime, paint and varnish both faces and all four edges.
- **F3, loose cells (C1).** Cells dropped into a bare steel box could short across their terminals on the walls or on each other. Fix: tape the terminals of every loose cell before it goes in and lay it in the 50 mm sand bed clear of the walls; a tape roll lives in the box.
- **F5, the phone in the tray (S7).** The camera bump had 2 mm of slack in its 34 mm window and the phone could turn 0.72 degrees between the lips: 5.7 mm of possible picture shift against a 5.3 mm margin. Fix: a 32 mm window (1 mm slack) and the phone pushed against the far lip, which leaves 3.3 mm with 0.3 degrees of tray tilt.

Method: STANDARDS section 20, fix rule. Calculations: `docs/04-calcs/stem_check.py`, results in `docs/04-calcs/stem_results.csv`; design values from `cad/src/model.py` and WML-CAL-001 (`docs/04-calcs/sizing.py`).

Claims checked

Table 1. Claims, governing science and verdicts. Rows that failed first show the verdict after the fix; the first verdict is in the check column.

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
S1	Lens 440 mm above the mat with a 26 mm-equivalent lens sees a 586 x 439 mm frame	Physics	Optics and radiation	Pinhole camera: frame = height x sensor size / focal length, on the 35 mm-equivalent 4:3 frame (34.62 x 25.96 mm)	claimed 439 mm (frame depth), estimate 439 mm (frame depth) (ratio 1.00)	Within known science	WML-CAL-001 section 6
S2	The frame stays on the 450 mm deep mat with the phone's placement in the tray allowed for	Physics	Optics and radiation	Pinhole camera: frame depth 439 mm at 26 mm; with 3.3 mm of placement stack (S7) the lens must be 25.76 mm equivalent or longer, 1 % under a quoted 26 mm that is itself rounded (independent review 2026-10-25). Lowering the lens to 420 mm gives 15.3 mm a side and fits lenses of 24.2 mm and longer: proposed to Amish as D2, since 440 mm was his decision (N3)	26 mm (35 mm-equivalent focal length) >= 25.76 mm (35 mm-equivalent focal length)	Plausible, to confirm at TRL 3	Open
S3	A visible-light camera does not reliably separate opaque HDPE from PP or identify carbon-black plastics;	Physics	Optics and radiation	Polymers are told apart by C-H overtone absorption in the short-wave infrared (Neo et al. 2022, as cited in WasteWise Scan WSC-PRB-001), not by visible colour; carbon black pigment prevents NIR sorting of	The design claims a limit, not a capability: 9 % of items (assumed stream) are routed to Unsure (WML-CAL-001 section 8)	Within known science	WML-CAL-001 section 8; PR-002 not met by design

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
	these go to Unsure			black packaging (WRAP)			
S4	An 8 h shift of 500 scans (tap-to-scan, screen asleep between scans, voice on, button linked) uses 30 % or less of a 4,000 mAh battery	Physics	Electromagnetism and circuits	Energy balance: 7.5 J a scan, 0.03 W asleep, 0.5 W for 1 s of voice, 0.005 W Bluetooth link; battery 15.4 Wh	9 % of battery <= 30 % of battery	Within known science	WML-CAL-001 section 5 (PR-010)
S5	The 10,000 mAh power bank puts about 26.7 Wh into the phone battery	Physics	Electromagnetism and circuits	Conservation of energy: 37 Wh x 0.85 boost x 0.85 charger	claimed 26.7 Wh, estimate 26.7 Wh (ratio 1.00)	Within known science	WML-CAL-001 section 5
S6	A firm 5 N tap on the screen moves the picture on the mat by less than 1 mm (one grid line's width)	Physics	Mechanics and statics	Cantilever bending of the 25 x 25 x 2 aluminium post and arm, E = 69 GPa (6063); bolted joints and clamps taken as rigid; the lens's own 0.12 mm drop is left out, which opposes the axis turn, so the figure is conservative	0.3 mm <= 1 mm	Within known science	WML-CAL-001 section 11; WML-BLD-001 first check 'Stand rigid'
S7	The phone's placement in the tray keeps the picture within the 5.3 mm margin	Engineering and technology	Manufacturing tolerances	A whole-stand tilt barely moves the picture (0.24 mm a degree: the lens moves one way, the axis turns the other); the phone does: 7.7 mm a degree of axis tilt,	3.3 mm <= 5.339 mm	Within known science	WML-DEC-001 F5 (fix implemented: tray window, BLD step 7, first check)

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
				5.1 mm a degree of yaw, plus bump slack in the window. First verdict: Fails, fix proposed (34 mm window and a free phone: 2.0 mm slack plus 0.72 deg yaw, 5.7 mm). Fix F5: 32 mm window (1 mm slack) and the phone pushed against the far lip (no yaw); with 0.3 deg of tray tilt (assumed) the stack is 3.3 mm			
S8	In shade on a 40 C Pune afternoon the phone, used tap-to-scan, stays below the 45 C Li-ion charge limit	Physics	Heat transfer	Steady heat balance, h = 10 W/(m2 K) over both faces (0.024 m2); Li-ion charge window 0 to 45 C (Battery University BU-410). In direct sun (about 10 W absorbed) the phone would reach about 85 C, so the rig stands in shade. With the screen on and charging (case B) the phone runs about 10 K over the air, so that fallback holds only below about 35 C (independent review 2026-10-25)	40.7 C <= 45 C	Within known science	WML-CAL-001 section 13; WML-BLD-001 safety stop S4 (shade); PR-020
C1	Loose cells put in the steel hazard box are	Chemistry	Electrochemistry and batteries	A cell whose terminals touch a conductor (the steel walls,	First verdict: Fails, fix proposed (a bare steel box with a	Within known science	WML-DEC-001 F3 (fix implemented:

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
	much less likely to short across their terminals			another cell, a can) discharges through it and heats; lithium cells can then go into thermal runaway. Taping terminals is the standard control (UCLA EHS, battery safety)	sand floor; terminals could touch the walls or each other). Fix F3: the terminals of every loose cell are covered with PVC tape before it goes in, and cells are laid in the 50 mm sand bed clear of the walls; a tape roll lives in the box		BOM line 5, PRC safety, PR-018)
C2	The 50 mm sand bed absorbs the heat of one phone-size cell in thermal runaway with a mean rise under 100 K	Chemistry	Physical chemistry	Heat balance: 15.4 Wh x 8 (assumed upper bound on total heat release against stored electrical energy) into 6.0 kg of sand at 830 J/(kg K): bulk heat capacity only; local temperature near the cell (hundreds of C), venting and spread to other cells are not assessed. The steel box keeps the fire from paper and film; sand does not stop a cell in runaway	89 K <= 100 K	Within known science	WML-CAL-001 section 14
C3	The painted hardboard mat stays flat when wet, dirty items lie on it	Chemistry	Materials and polymer chemistry	Hardboard takes up water through unsealed faces and edges: 24 h thickness swelling up to 25 % for 3.2 mm board (EN 317 rating, Moelven hardboard datasheet); a	First verdict: Fails, fix proposed (the build plan painted and varnished the top face only). Fix F2: prime, paint and varnish both faces and all four edges, so the	Plausible, to confirm at TRL 3	Open

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
				coating on one face only also cups the board as the other face swells	board is sealed all over and the coatings balance. That it then stays flat is not shown: chips and scratches let water in (independent review 2026-10-25); a 24 h wet-cloth soak and cup measurement at TRL 4 confirms it		
C4	The model never certifies chemical content, food-contact suitability or hazardous-waste status	Chemistry	Analytical	A photo records reflected visible light; it carries no chemical analysis	Scope claim (PRB out of scope; PRC safety)	Within known science	WML-PRB-001 out of scope
B1	The user triggers each scan without repeated reaching at shoulder height	Biology	Ergonomics	Two-link arm (Drillis and Contini proportions, Winter 2009) for the 5th percentile Indian woman (stature 1,356 mm, Nag, Nag and Desai 2003); ISO 11226 screens upper arm elevation above 60 deg for static postures (a tap a minute is not static; the 59 deg moves to 51 or 68 deg with the shoulder 100 or 200 mm behind the bench edge)	First verdict: Plausible (tapping the screen needs 59 deg upper arm elevation at about one scan a minute, at the 60 deg screen). Fix F1: a Bluetooth shutter button on the bench edge, 45 mm in from the user's edge at bench height, takes the photo; no reach to the phone	Within known science	WML-DEC-001 F1 (fix implemented: BOM line 12, model, BLD, PR-015)

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
B2	A user at the 5th percentile can read the result (a 40 mm bin number) on the screen lying flat at 1,215 mm	Biology	Ergonomics	Visual angle of the foreshortened symbol: the 5th percentile user's eye is 54 mm above the screen and 415 mm back, 82.6 deg off the screen normal; a 10 mm icon subtends 11 arcmin against the 16 arcmin minimum of ISO 9241-303 (first verdict: Fails, fix proposed). Fix F1: the result fills the screen in the class colour with the bin number at least 40 mm tall, and the phone says the bin number aloud by default. ISO 9241-303 assumes a near-normal view; at 83 deg the digit is squashed about 7.7 to 1 and glare and contrast loss are likely, so legibility is confirmed by a trial with a short user (independent review 2026-10-25); colour and voice do not depend on it	43 arcmin >= 16 arcmin	Plausible, to confirm at TRL 3	Open
B3	The rig adds little exposure beyond the existing handling of	Biology	Toxicology	Cut-resistant gloves, no handling of items closer to the camera, hazards	Safety claim only	Within known science	WML-PRC-001 safety; WML-BLD-001 safety stops

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
	sharps and dirty items, apart from the stored cells of C1 and C2			to the box (PRC safety)			
M1	The 8-bit model file is about 4.4 MB	Mathematics	Units and dimensional analysis	Parameters x 1 byte + 5 % file overhead (MobileNetV3-Large, 5.4 M parameters less the ImageNet classifier plus two heads)	claimed 4.4 MB, estimate 4.36 MB (ratio 1.01)	Within known science	WML-CAL-001 section 2 (PR-006)
M2	Photo to result in 300 ms or less on a low-cost phone	Mathematics	Orders of magnitude	217 M multiply-accumulates at 1.2 to 4.9 GMAC/s (TensorFlow Models, MobileNet README: 44 ms on a Pixel 1) plus preprocessing	201 ms at the slow end from a preview-size frame (20 ms preprocessing); 331 ms if a full 12 MP frame is decoded first, which the independent review (2026-10-25) put at 100 to 300 ms on an A53-class phone. Met only if the app classifies a preview-size frame; to measure	Plausible, to confirm at TRL 3	Open
M3	400 hazard items with 3 misses or fewer show hazard recall of at least 98 % at 95 % confidence, for the evaluation population	Mathematics	Statistics and uncertainty	One-sided Clopper-Pearson lower bound; valid for an independent sample of the evaluation population only: hazards collected on purpose and photos from one site are not that, so field recall can	0.9807 recall >= 0.98 recall	Within known science	WML-CAL-001 section 9 (PR-003, PR-011)

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
				be lower; report each hazard type separately			
M4	About 230 field images per label give 90 % accuracy on answered items at 70 % coverage	Mathematics	Statistics and uncertainty	Power-law learning curve, error = 0.03 + a n^-0.5 anchored at 25 % for 100 images (assumed); power laws fit deep-learning learning curves (Hestness et al. 2017), but the constants for field waste photos are unknown	128 to 591 images per label across the sensitivity cases; the design figure of 300 covers six of the nine cases. The review recalls a scaling exponent nearer 0.3 for ImageNet (Hestness et al. 2017, to check), which would need about 330	Plausible, to confirm at TRL 3	Open
M5	About 66 % of items get an answer (63 % graded, 3 % true hazard flags) at 90 % grade accuracy	Mathematics	Statistics and uncertainty	Selective classification: abstaining on the least-confident share u removes 2u of the errors (assumed); stream composition assumed	Corrected after the independent review (false hazard alarms leave the graded pool): 66 %, 4 points under PR-004; rests on the assumed stream, enrichment factor and perfect abstention on camera-blind items	Plausible, to confirm at TRL 3	Open
M6	About 1,065 answered items estimate a 90 % accuracy to within 2 points (95 %)	Mathematics	Statistics and uncertainty	Normal approximation to the binomial, half-width 1.96 sqrt(p(1-p)/n)	1.8 points <= 2 points	Within known science	WML-CAL-001 section 9
M7	Labeling 10,000 photos	Mathematics	Orders of magnitude	Photos x 20 s x 1.3 review	claimed 72.2 h, estimate 72.2 h (ratio 1.00)	Within known science	WML-CAL-001 section 7

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
	takes about 72 h (labeling only; capture and NIR ground truth for camera-blind grades excluded)						
E1	A phone image classifier can tell broad material classes apart from one photo	Engineering and technology	Sensing resolution	TrashNet: six classes of single items photographed with phones on a plain board were told apart (Thung and Yang 2016; the reported accuracy is to recheck against the CS 229 report, independent review 2026-10-25); a clean board is a weak proxy for the field	The principle is shown on clean photos; field accuracy by grade is PR-001 and PR-002 (at risk) and M4	Within known science	WML-PRB-001 prior work
E2	At full resolution a molded 6 mm resin code digit spans at least 20 pixels (pixel count only)	Engineering and technology	Sensing resolution	Pixel pitch on the mat = frame width / 4,000 px; 20 px a character is the design figure (assumed). Legibility of embossed, same-colour codes also depends on lens sharpness and light, not shown here	41 px >= 20 px	Within known science	WML-CAL-001 section 6
E3	The model and its activations need about 8 MB of the phone's 3	Engineering and technology	Power and link budgets	Weights plus the two largest activation tensors (112 x 112 x 64, 8-bit, double-buffered)	claimed 8 MB, estimate 7.61 MB (ratio 1.05)	Within known science	WML-CAL-001 section 3

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
	GB of memory (camera frame buffers, about 48 MB at 12 MP, come on top)						
E4	The shutter button's CR2032 cell lasts at least three months of shifts	Engineering and technology	Power and link budgets	225 mAh at 0.05 mA mean while linked (assumed, no datasheet); cheap buttons may draw 0.1 to 0.5 mA, and above about 0.45 mA the cell fails the 78-shift screen (independent review 2026-10-25)	562 shifts at 0.05 mA; 62 at 0.45 mA. The screen tap stays as the fallback; measure the bought button	Plausible, to confirm at TRL 3	Open
OB1	Option B: the wide camera 450 mm over the tray floor sees the whole 600 x 400 mm tray	Physics	Optics and radiation	Frame = 2 h tan(FOV / 2), 102 x 67 deg (Camera Module 3 Wide)	596 mm (frame depth) >= 400 mm (frame depth)	Within known science	WML-CAL-001 section 15
OB2	Option B: a 40 mm bottle cap spans at least 16 px at a 1,280 px detector input	Engineering and technology	Sensing resolution	Pixel pitch = frame width / 1,280 px; small-object detectors need about 16 to 32 px (assumed); an 11 mm button cell is only about 13 px	46 px >= 16 px	Within known science	WML-CAL-001 section 15
OB3	Option B: a 20,000 mAh power bank runs the Pi 5 and	Physics	Electromagnetism and circuits	Energy balance: Pi 5 idle 2.7 W, display 1.5 W, detector bursts 6 W for 3 s	34.1 Wh <= 62.9 Wh	Within known science	WML-CAL-001 section 15

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
	display for an 8 h shift			(assumed); 74 Wh x 0.85 boost			
OB4	Option B: the detector answers a 1,280 px tray photo in about 1 s on the Pi 5	Mathematics	Orders of magnitude	YOLOv8n at 640 px is 8.7 GFLOP (Ultralytics); x4 at 1,280 px; 20 to 40 GFLOP/s effective on the Pi 5 (assumed)	claimed 1 s, estimate 1.16 s (ratio 0.86)	Within known science	WML-CAL-001 section 15
OB5	Option B: hazard recall of 98 % holds for items at least half visible in a tray of 10 to 20	Mathematics	Statistics and uncertainty	Detection recall falls for occluded objects; no field data for waste trays	Untested; would need its own field evaluation set	Plausible, to confirm at TRL 3	Open

Verdicts: **Within known science**, **Plausible, to confirm at TRL 3**, **Fails, fix proposed**, **Fails**. No row is Fails.

Fields not relevant

Every field has at least one row. Within them, these branches are not relevant:

- Physics, fluid mechanics, acoustics and nuclear: nothing flows, the only sound is the phone's voice prompt, and nothing is radioactive.
- Chemistry, organic and environmental chemistry: the station handles waste but changes none of it chemically; disposal routes are the partner's and the municipality's (PRB).
- Biology, ecology, agronomy and epidemiology: no living material is grown or released; biological contamination of dirty items is a handling hazard (B3).

Concept options

- **Option A, phone stand over a mat (current design):** 26 rows. Within known science after fixes F1 to F3 and F5. Open: S2 (lens, D2 to Amish), M2, M4, M5 (field or bench measurements), B2, C3, E4 (trials on bought or made parts).
- **Option B, tray detector:** five rows of its own (OB1 to OB5) plus the shared physics of the station. Within known science on optics, resolution, power and speed; OB5 (hazard recall for items at least half visible in a crowded tray) is Plausible and would need its own field evaluation. Not dropped.

Independent review

A second agent (Claude, run headless with `claude -p`, 2026-10-25) saw only this document, WML-CAL-001, WML-PRS-001 and `docs/04-calcs/sizing.py`, and was asked to refute the headline results and every "Within known science" verdict. It recomputed and could not refute S1, S4, S5, S6, S8, M1, M3 (arithmetic), M6, M7, E2, E4 (arithmetic), OB1, OB3, OB4 and the ergonomic angles of B1 and B2. It found, and this session changed:

1. **S7 (stand tilt): refuted.** A rigid stand tilting about its board barely moves the picture (0.24 mm a degree, not 15.1); the phone's own tilt (7.7 mm a degree), yaw (5.1 mm a degree) and slack in the window dominate. The first pass stacked to 5.7 mm against 5.3 mm. Fixed under the fix rule (F5) and restated.
2. **S2, PR-019 (5.3 mm margin): weakened.** The margin is 1.2 % of the depth, inside the rounding of a quoted 26 mm lens. S2 is now Plausible and open, PR-019 at risk, and lowering the lens to 420 mm is proposed to Amish (D2).
3. **M2, PR-005 (preprocessing): weakened.** Decoding a full 12 MP frame on an A53-class phone takes 100 to 300 ms, not 20 ms. Photo to result is 64 to 201 ms from a preview-size frame and up to 331 ms from a full frame; M2 is Plausible and PR-005 at risk until the app's capture path is measured.
4. **S4, PR-007 (sleeping screen): weakened.** A 0.12 W standby gives 14 % of the battery, still within 30 %. Waking the phone and opening the camera adds 0.5 to 1.5 s, so the worst case of PR-007 is 3.6 s; and whether a Bluetooth button reaches the app with the screen asleep is unproven (DEC confirm item 5).
5. **S8 against the power-bank fallback: contradiction found.** With the screen on and charging, the phone runs about 10 K over the air, so case B of PR-010 holds only below about 35 C. Stated in CAL sections 5 and 13.
6. **M5, PR-004 (coverage): corrected.** False hazard alarms leave the graded pool: 66 % answered, not 68 %.
7. **M4 (learning curve):** the design figure covers six of nine sensitivity cases, not "the central cases"; the reviewer recalls a scaling exponent nearer 0.3 for ImageNet, to check.
8. **M3:** the bound holds for the evaluation population only; hazards are to be reported by type. **M6:** stratified, clustered sampling widens the 1.8 points. **M7:** labeling only, capture and NIR ground truth excluded. Wording changed.
9. **B1, B2, B3, C1, C2, C3, E1, E2, E3, E4, S3, S5, S6, OB2, OB5:** wording narrowed (ISO 11226 is for static postures; legibility at 83 degrees needs a trial; the rig adds the stored-cell hazard; tape makes shorts much less likely, not impossible; sand gives bulk heat capacity only; sealing does not prove flatness; TrashNet's accuracy to recheck; pixels are not legibility; camera buffers come on top of the 8 MB; the button's current is an assumption; S5's check text now tests 26.7 Wh). B2, C3 and E4 are now Plausible and open.
10. **Documents:** the PRS list of requirements at risk now matches the statuses; PR-013's license review is to include the ImageNet pretraining weights (noted in WML-REVIEW).

Sources

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- ISO 9241-303:2011. *Ergonomics of human-system interaction, Part 303: Requirements for electronic visual displays* (character height 16 arcmin minimum, 20 to 22 preferred). Standard not purchased; values as widely reported, to be confirmed against the text.
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- UCLA Environment, Health and Safety. "Battery Safety" (taping terminals prevents short circuits, especially with lithium batteries). https://ehs.ucla.edu/news/battery-safety. Checked 2026-10-25.
- Ultralytics. YOLOv8 model table (YOLOv8n, 8.7 GFLOP at 640 px). https://docs.ultralytics.com/models/yolov8/. Not rechecked in this session.
- WRAP. Recyclability of black plastic packaging (carbon black prevents NIR sorting), cited through WasteWise Scan WSC-PRB-001.
- WML-CAL-001 (`docs/04-calcs/01-sizing.md`) for every design number.